

# Zhiming Huang

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| EMPLOYMENT                    | <b>Postdoc</b> at Paul G. Allen School of Computer Science & Engineering<br>▪ <b>University of Washington</b> , WA, USA<br>▪ Advisor: Prof. Jamie Morgenstern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Dec 2025 – Now                                                             |
| EDUCATION                     | <b>University of Victoria</b> , BC, Canada<br>▪ <b>Ph.D.</b> in Computer Science (Aug 2020 – Aug 2025)<br>▪ Advisor: Prof. Jianping Pan<br>▪ Dissertation: <a href="#">Swap-Regret-Minimizing Bandits for Distributed Network Optimization</a><br>▪ <b>M.Sc.</b> in Computer Science (Sep 2018 – Aug 2020)<br>▪ Advisor: Prof. Jianping Pan<br>▪ Thesis: Thompson sampling-based online decision making in network routing<br><b>B.Eng.</b> in Communication Engineering<br>▪ <b>Northwestern Polytechnical University</b> , Shaanxi, China                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Sep 2018 – Aug 2025<br><br><br><br><br><br><br><br><br>Sep 2014 – Jun 2018 |
| RESEARCH INTERESTS AND SKILLS | ▪ Learning-theoretic foundations of networking and distributed systems, including online learning and bandit algorithms for congestion control, network selection, and resource allocation, with provable performance guarantees.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                            |
| PUBLICATIONS                  | <b>CONFERENCE PUBLICATIONS</b><br><br>[1] <b>Z. Huang</b> , B. Hu and J. Pan, Bridging the Regret Gap in Combinatorial Thompson Sampling: Worst-Case Guarantees and Algorithmic Refinement, <i>IEEE International Conference on Computer Communications (INFOCOM)</i> , May 2026.<br>[2] Q. Zhang, <b>Z. Huang</b> , J. Zhao, A. Ahangarpour and J. Pan, Faster Exploration and Exploitation for Communication Environment Awareness in Starlink, <i>IEEE International Conference on Computer Communications (INFOCOM)</i> , May 2026.<br>[3] <b>Z. Huang</b> and J. Pan, Faster Convergence for Unknown-game Bandits, <i>IEEE International Conference on Computer Communications (INFOCOM)</i> , May 2025.<br>[4] <b>Z. Huang</b> and J. Pan, Adversarial Semi-Bandits with Moving Arms, <i>IEEE International Conference on Computer Communications (INFOCOM)</i> , May 2025.<br>[5] B. Hu, <b>Z. Huang</b> , T. Zhang, M. Lécuyer and N. Hegde, Connecting Thompson Sampling and UCB: Towards More Efficient Trade-offs Between Privacy and Regret, <i>International Conference on Machine Learning (ICML)</i> , May 2025.<br>[6] Q. Zhang, <b>Z. Huang</b> , J. Zhao, and J. Pan, A Congestion Control Test Suite for Real-Time Communication, <i>ACM Multimedia Systems Conference (MMSys)</i> , Apr 2025.<br>[7] <b>Z. Huang</b> , B. Hu, and J. Pan, Gaussian Randomized Exploration for Semi-bandits with Sleeping Arms, <i>NeurIPS 2024 Workshop on Bayesian Decision-making and Uncertainty</i> , Dec 2024.<br>[8] F. Amri, <b>Z. Huang</b> , K. Liu, and J. Pan, Energy-aware Inter-data Center VM Migration over Elastic Optical Networks, <i>IEEE International Conference on Communication (Globecom)</i> , Dec 2023.<br>[9] <b>Z. Huang</b> and J. Pan, A Near-optimal High-probability Swap-Regret Upper Bound for Multi-agent Bandits in Unknown General-sum Games, <i>Uncertainty of Artificial Intelligence (UAI)</i> , Aug 2023.<br>[10] <b>Z. Huang</b> , K. Liu and J. Pan, End-to-end Congestion Control as Learning for Unknown Games, <i>IEEE International Conference on Distributed Computing Systems (ICDCS)</i> , Jul 2023.<br>[11] <b>Z. Huang</b> , B. Hu, and J. Pan, Poster: Multi-agent Combinatorial Bandits with Moving Arms, <i>IEEE International Conference on Distributed Computing Systems (ICDCS)</i> , Jul 2021.<br>[12] B. Hu, Y. Chen, <b>Z. Huang</b> , N. Mehta, and J. Pan, Intelligent Caching Algorithms in Heterogeneous Wireless Networks with Uncertainty, <i>IEEE International Conference on Distributed Computing Systems (ICDCS)</i> , Jul 2019. |                                                                            |

- [13] R. Liu, J. Song, **Z. Huang**, and J. Pan, EQRC: An Enhanced QR Code-based Secure E-coupon Transaction Framework, *IEEE International Conference on Communication (ICC)*, May 2019.

#### JOURNAL PUBLICATIONS

- [1] W. Yang, L. Cai, S. Shu, A. Sepahi, **Z. Huang** and J. Pan, QoS-driven Contextual MAB for MPQUIC Supporting Video Streaming in Mobile Networks, *IEEE Transactions on Mobile Computing (TMC)*, Nov 2024.
- [2] **Z. Huang** and J. Pan, Game-theoretic Bandits for Network Optimization with High-probability Swap-regret Upper Bounds, *IEEE/ACM Transactions on Networking (TON)*, Jul 2024.
- [3] B. Hu, **Z. Huang**, N. Mehta, and N. Hegde, Near-Optimal Algorithms for Differentially Private Online Learning in a Stochastic Environment, *Arxiv*, Jul 2024.
- [4] K. Liu, J. Wang, **Z. Huang**, and J. Pan, Sampling-based Job Placement and Load Balancing for Heterogeneous Deep Learning Clusters, *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, Apr 2024.
- [5] **Z. Huang** and J. Pan, Distributed Learning of Unknown Games for HetNet Selection, *IEEE Transactions on Network Science and Engineering (TNSE)*, Jan 2024.
- [6] **Z. Huang**, Y. Xu, and J. Pan, TSOR: Thompson Sampling-based Opportunistic Routing, *IEEE Transactions on Wireless Communications (TWC)*, vol. 20, no. 11, pp. 7272-7285, May 2021.
- [7] **Z. Huang**, B. Hu, and J. Pan, Caching by User Preference with Delayed Feedback for Heterogeneous Cellular Networks, *IEEE Transactions on Wireless Communications (TWC)*, vol. 20, no. 3, pp. 1655-1667, Mar 2021.
- [8] R. Liu, J. Song, **Z. Huang**, and J. Pan, EQRC: A Secure QR Code-based E-coupon Framework Supporting Online and Offline Transactions, *Journal of Computer Security (JCS)*, vol. 28, no. 5, pp. 577-605, Sep 2020.
- [9] **Z. Huang**, R. Zhang, J. Pan, Y. Jiang, and D. Zhai, A Framework of Multipath Clustering based on Space Transformed Fuzzy c-Means and Data Fusion for Radio Channel Modeling, *IEEE Transactions on Vehicular Technology (TVT)*, vol. 69, no. 1, pp. 4-15, Jan 2019.

#### TEACHING EXPERIENCE

##### INSTRUCTOR

- **CSC111: Fundamentals of Programming with Engineering Applications**

Spring 2024

##### TEACHING ASSISTANT

- **Lab and Tutorial Instructor for CSC361: Computer Communication and Networks**
- **Tutorial Instructor for CSC360: Introduction to Operating Systems**

#### PROFESSIONAL ACTIVITIES

##### PROGRAM COMMITTEE

- Shadow TPC member for ACM CoNEXT 2025

##### REVIEWER

- IEEE Transactions on Wireless Communications
- IEEE Transactions on Intelligent Transportation Systems
- IEEE Transactions on Artificial Intelligence
- IEEE Transactions on Signal Processing

#### AWARDS & SCHOLARSHIPS

- **Canada NSERC Postdoctoral Fellowship** 2025
- **IEEE INFOCOM 2025 Travel Award** 2025
- **British Columbia Graduate Scholarship** 2024
- **UVic President's Fellowship in Research-enriched Teaching** 2023
- **ICDCS 2023 Travel Award** 2023
- **UAI 2023 Scholarships** 2023
- **Melva J. Hanson Graduate Scholarship** 2022
- **IEEE ICDCS 2021 Best Poster Award** 2021
- **Howard E. Petch Research Scholarship** 2020